DECISION SCIENCES INSTITUTE

An AI-driven Transfer Student Advising System: Design, Implementation, and Evaluation

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ABSTRACT

Transfer students face advising challenges that conventional systems do not address well: records are split across institutions, and predictive models are ~~typically~~ trained on students who entered as freshmen. We present ACOSUS, an AI-driven advising ~~platform~~ prototype designed ~~specifically~~ for transfer students. The system ~~incorporates~~ combines a dual-survey approach that separates outcome measurement from predictor ~~collection and~~ collection, a progressive learning framework ~~suitable~~ for institutions with limited ~~data.~~ data, and an advisor dashboard for structured transfer-specific information. A pilot study found positive ~~participant~~ ease-of-use ratings ~~of~~ and moderate perceived ~~usefulness and ease of use,~~ usefulness, supporting the design feasibility of the ~~system~~ user interface and ~~its~~ survey workflow. ~~The~~ Predictive ~~platform~~ validity ~~is now deployed and collecting data~~ remains for future longitudinal evaluation.

KEYWORDS: transfer students, intelligent advising, machine learning, higher education, student success prediction

INTRODUCTION

Many students begin at community colleges and later transfer to four-year universities to complete their degrees. National studies estimate that more than one-third of undergraduates follow this route (Monaghan & Attewell, 2015), and the proportion is even higher in computing and other STEM majors and among first-generation, low-income, and racially minoritized students (Ty et al., 2024; Wali & Hooshangi, 2025). Although this pathway offers a more affordable route to a bachelor's degree (Wang et al., 2023), it also creates coordination problems between sending and receiving institutions (P. M. Nguyen et al., 2025). Advisors must work with articulation agreements, uneven credit recognition, and limited access to students' pre-transfer coursework (D. V. Nguyen et al., 2024b; Roksa & Keith, 2008). For students from groups already underrepresented in STEM, these academic challenges often overlap with financial pressure and a weaker sense of belonging in a new institutional setting (P. M. Nguyen et al., 2025; Ty et al., 2024).

Persistence and timely graduation remain major concerns. Transfer students are more likely than native students to experience "transfer shock," a decline in GPA and sense of belonging during the first two semesters after matriculation (Jaggars et al., 2025), and they show higher attrition rates overall. Quantitative studies identify several measurable factors associated with retention and completion, including cumulative GPA, accepted credits, standardized test scores, distance between institutions, and financial aid (Jaggars et al., 2025; Roksa & Keith, 2008; Wang et al., 2023). These variables are useful, but they do not capture the full transfer experience (P. M. Nguyen et al., 2025). Qualitative work repeatedly points to social support, faculty mentorship, and campus engagement opportunities such as research assistantships and hackathons, all of which are difficult to represent in standard student-information systems (Thiry et al., 2023).

Advising is consistently identified as an important factor in transfer-student success (Maliszewski Lukszo & Hayes, 2020; Thiry et al., 2023). Students want clear, timely, and individualized guidance, but large caseloads leave many advisors with limited time to understand each student's background, goals, and constraints (Allen et al., 2014; P. M. Nguyen et al., 2025). Existing advising dashboards mostly reproduce registrar data and early-alert measures developed for native cohorts; they rarely include pre-transfer trajectories, articulation details, or information students typically share in advising conversations (D. V. Nguyen et al., 2024b; P. M. Nguyen et al., 2025). As a result, predictive models may misclassify student risk and reproduce existing inequities (Ty et al., 2024; Vargas et al., 2024). What is needed is an advising approach that combines administrative data with information about how transfer students actually experience the move between institutions (P. M. Nguyen et al., 2025).

To address this gap, we designed and evaluated an intelligent advising system that combines academic records with social, financial, and experiential data collected before and after transfer. This is primarily a design paper; the pilot study provides evidence that the design works in controlled settings, while large-scale predictive evaluation is deferred to future work. Our contributions are as follows:

1. We built an advising platform for transfer students that combines academic records with student-provided background information to create a unified longitudinal profile.
2. We developed a dual-survey architecture that separates outcome measurement (Target Surveys) from predictor collection (Factor Surveys), allowing consistent capture of transfer-specific ~~variables~~ variables. ~~while~~
3. ~~providing~~ We ~~advisors~~ implemented ~~with~~ a structured data-collection workflow and advisor dashboard that organize academic and nonacademic survey responses into student ~~profiles~~ profiles. ~~through~~ Formal ~~one~~ evaluation ~~dashboard.~~ of advisor usability remains future work.
4. We implemented a modular AI prediction architecture with configuration support for survey linkage, phase transitions, and hyperparameters, so researchers can adapt the system to different institutional settings without changing code.
5. We evaluated the student-facing system design in a controlled pilot study with transfer students, providing initial evidence that the ~~platform~~ interface, ~~operates~~ survey workflow and system operate as intended and ~~is~~ are perceived as useful, easy to use, and supportive of self-advising.

The remainder of the paper is organized as follows. LITERATURE REVIEW discusses related work on transfer-student success and intelligent advising. METHODOLOGY AND IMPLEMENTATION describes our system design and methodology, as well as implementation details. EVALUATION reports empirical findings from our pilot study. The last section concludes and outlines directions for future research.

LITERATURE REVIEW

Research on transfer-student outcomes can be grouped into two related areas (Wang et al., 2023). The first examines the personal and academic characteristics that shape transfer experiences. The second examines institutional practices, especially advising, that influence those outcomes.

Early models, including Ty et al. (2024)'s use of Social-Cognitive Career Theory, describe transfer success in terms of "person inputs" (e.g., socioeconomic status and self-efficacy) and "transfer receptivity," or the extent to which the receiving institution is prepared to support transfer students. Studies across STEM programs show that self-efficacy and goal-setting often improve after transfer, even as students adjust to unfamiliar academic expectations (Vargas et al., 2024). Factor-analysis studies further show that financial considerations, institutional reputation, distance from home, and family expectations shape students' decisions to transfer (Wang et al., 2023). Regression analyses connect post-transfer GPA to race/ethnicity and first-generation status (Ty et al., 2024). Topic modeling of open-ended survey responses and social-media discussions adds further detail, especially around frustration with credit evaluation, commuting, and limited opportunities for campus engagement (Standfast et al., 2024).

Advising research often uses the idea of "transfer student capital" to describe the knowledge and networks students develop while moving across institutions (Maliszewski Lukszo & Hayes, 2020). When formal advising falls short, many students turn to Reddit, Discord, or peer forums for help with financial aid, housing, and course selection (Standfast et al., 2024). Qualitative interviews show that relationship-based advising can ease transfer shock and strengthen students' sense of belonging (Thiry et al., 2023). At the same time, advisors report difficulty keeping up with changing articulation rules and program maps while managing large caseloads (D. V. Nguyen et al., 2024b; P. M. Nguyen et al., 2025).

Technology tools for advising have begun to appear, but most were built for native student populations (D. V. Nguyen et al., 2024b). These systems usually rely on registrar data and LMS (Learning Management Systems) clickstreams, and they often miss transfer-specific variables such as credit-transfer friction, completion patterns, and belonging-related indicators (D. V. Nguyen et al., 2024b; P. M. Nguyen et al., 2025). Prior studies warn that models trained on narrow student populations can produce biased predictions and disadvantage students from underrepresented groups (Ty et al., 2024; Vargas et al., 2024). Only a small number of prototype systems focus specifically on transfer students, and most remain at the pilot stage (MacDonald et al., 2024).

Transfer-oriented technologies address different parts of the advising process. ASSIST provides official articulation and transferability information for California public institutions and participating Association of Independent California Colleges and Universities institutions +(ASSIST, 2026). Transferology allows students to explore how courses, standardized exams, and military credits may transfer to participating institutions, while TES supports staff who research transfer credit, route evaluations, manage equivalencies, and publish results +(CollegeSource, 2026a, 2026b). Navigate360 documents a broader institutional platform with student profiles, surveys, advising, alerts, and embedded AI workflows +(EAB, 2026). Table +1 compares these documented purposes with ACOSUS. These purposes suggest that the systems are complementary rather than interchangeable.

Research prototypes complement these operational tools by exploring planning and advising tasks. Examples include hypothetical transfer-plan optimization, decision-theoretic course planning based on student goals and performance models, and transcript- and degree-plan-aware course recommendation (D. V. Nguyen et al., 2024a, pp. 3, 5, 12, 19; Mattei et al., 2014, pp. 1, 4; Mi et al., 2025, pp. 1, 4–6). These studies evaluate transfer-plan construction, advising recommendations and explanations, and context-aware course recommendations, while the present ACOSUS pilot focuses on the feasibility of its transfer-specific survey and interface workflow.

Researchers have therefore called for mixed-methods approaches that combine quantitative data with qualitative evidence (P. M. Nguyen et al., 2025). Topic modeling is one useful approach

Table 1: Transfer-specific comparison of documented system purposes.

System	Documented purpose	Documented capability	Documented scope relative to ACOSUS
ASSIST	Official articulation and transferability repository for California public institutions and participating independent institutions	Authoritative course, general-education, and major-agreement information within its covered system	The cited source documents articulation information, not broader advising-context collection
Transferology	Student-facing exploration of how courses, standardized exams, and military credits may transfer	Transfer-credit results based on courses and other credit entered by the student	The cited workflow depends on participating-institution evaluations and centers on transfer-credit decisions
TES	Staff-facing research, evaluation routing, equivalency management, and publication	Documented workflow for institutional transfer-credit review and communication	The cited source documents articulation and equivalency workflows, not transfer-experience data collection
Navigate360	Institution-wide student-success CRM with profiles, surveys, advising, alerts, and AI-assisted workflows	Combines institutional and engagement signals across student and staff workflows	The cited page does not document a dual-survey architecture that separates readiness measurement from predictor collection
ACOSUS	Dual-survey architecture that separates readiness measurement from predictor collection, with advisor-visible profiles and an exploratory AI path	Structures academic and nonacademic transfer context while keeping Target and Factor Surveys separate	Does not evaluate or determine course-transfer equivalencies; advisor usability and predictive validity have not been established.

because it can identify recurring themes in narrative responses that do not appear in structured records. Deng et al. (2021) applied Revealed Causal Mapping to survey narratives to examine how social media use shapes the academic experiences and social capital of first-generation students, and Thiry et al. (2023) used mixed-methods case studies to develop support strategies for STEM transfer students. Even so, few studies have incorporated these kinds of unstructured findings into automated advising systems. This points to a broader systems problem: even when researchers identify the right factors, institutions often do not have a reliable way to collect, preserve, and reuse that information for advising or prediction.

Data scarcity follows directly from that limitation. The issue is not a lack of transfer students, but the fact that their records are split across institutions (Roksa & Keith, 2008). When students transfer, only part of the academic record usually moves with them: credits earned, GPA, and course equivalencies. The broader context that community college advisors often know (family obligations, financial stress, career goals, and early warning signs) rarely follows the student to the receiving institution (D. V. Nguyen et al., 2024b). Much of this information remains undocumented or stored in systems that the receiving university cannot access (Allen et al., 2014). Advisors then have to reconstruct that context through time-consuming intake conversations. Prediction systems face the same problem: each institution must build its own labeled dataset over time, one student at a time (Coleman et al., 2019). This cold-start problem is especially serious in small-cohort settings, where the factors most predictive of success are often the least available in institutional records (Coleman et al., 2019).

Our work addresses these gaps from a systems perspective. We combine standard academic records (e.g., transfer credits, GPA history, and enrollment status) with information students provide about their own experiences. The system uses these data to support advising through a lightweight, interpretable prediction process and a dashboard that organizes transfer-relevant information in one place. Early pilot feedback suggests that this approach addresses a practical need in transfer-student support and merits further evaluation at larger scale. The pilot evaluated the student-facing workflow; prediction quality and advisor use require separate study.

METHODOLOGY AND IMPLEMENTATION

This section presents the design and implementation of ACOSUS. We first describe the overall system architecture, then the dual-survey data collection approach, the processing of survey responses, the progressive learning framework for small-data settings, and the configuration options that support institutional adaptation.

System Architecture

ACOSUS uses a four-layer architecture that separates presentation, business logic, data storage, and predictive modeling (Figure 1). This structure allows each layer to scale independently and supports both student-facing prediction and advisor-facing information management.

The **Presentation Layer** provides role-specific interfaces through a web application. Students complete surveys, view available predictions, and submit feedback. Advisors use a dashboard that organizes student profiles by the same categories used in the Factor Surveys, which reduces the need to assemble information from multiple sources. Administrators configure survey instruments, trigger supported model training, and monitor system activity.

The **Business Logic Layer** manages application workflows and ~~controls~~ is designed to control transitions between phases of the progressive learning framework. It determines which interface the student sees at a given time: data collection only during the foundation phase, and prediction with feedback in later phases.

The **Data Layer** stores users, surveys, responses, and model metadata in a document-oriented format. This structure allows researchers to revise survey instruments without repeated database migrations.

The **Model Layer** runs as a separate service for data normalization, prediction across all three pipeline stages, and model training. The prototype includes a transparent priority-weighted readiness calculation and supports exploratory similarity-based modeling. Asynchronous job queues keep training tasks from blocking prediction requests. Model versioning ensures that updated models are deployed only after they perform better on held-out real student data.

Table +2 distinguishes what the pilot evaluated, what the broader prototype supports, and what remains aspirational.

Table 2: Implementation and evaluation status at revision time.

Status	**Functions**	**Evidence boundary**
Implemented and pilot-evaluated	Target and Factor Survey workflow; structured response collection	Student pilot evaluated the survey workflow and perceptions; no predictions were shown
Implemented beyond the pilot scope	Advisor dashboard and profiles; deterministic readiness measure; exploratory modeling path	Available in the prototype; not evaluated in the student pilot or in a formal advisor study
Future work	Synthetic augmentation; advanced models; automatic stage transitions; longitudinal outcome validation; fairness analysis; multi-institutional validation	Proposed only; recruiting participants from six universities does not constitute institutional validation

This architecture defines where the major system functions reside, but it does not yet explain how transfer-specific information enters the platform. We address that next through the dual-survey design.

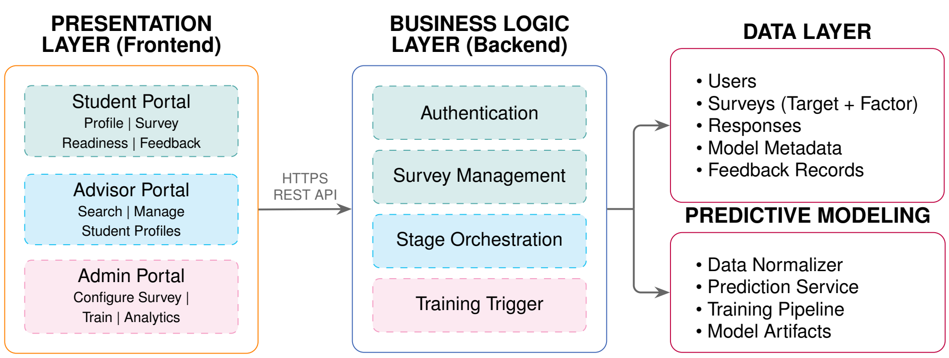

Figure 1: ACOSUS system architecture. Four layers separate presentation, business logic, data persistence, and predictive modeling, enabling independent scaling and clear separation of responsibilities.

Diagram text changes: Authentication; Survey Management; Stage Orchestration; Training Trigger; Student Portal [-1pt] Profile | Survey [-1pt] ~~Prediction~~ Readiness | Feedback; Advisor Portal [-1pt] Search | Manage [-1pt] Student Profiles; Admin Portal [-1pt] Configure Survey | [-1pt] Train | Analytics

Dual-Survey Architecture

~~This~~ In ~~section~~ this ~~introduces~~ paper, **student success** refers to persistence after transfer, satisfactory academic progress and credit accumulation, and eventual bachelor's degree completion. These outcomes require longitudinal institutional records and were not observed in the ~~terms~~ pilot. ~~used~~ The ~~throughout~~ Target ~~the~~ Survey ~~paper.~~ instead ~~ACOSUS~~ provides ~~treats~~ a ~~transfer~~ proximal, self-reported **readiness measure** based on students' reports of academic confidence, commitment, time management, and related constructs at one point in time. The measure is not an observed success outcome or evidence that a student ~~success~~ will ~~prediction~~ persist ~~as~~ or ~~a supervised learning problem in which an outcome is predicted from a set of input variables. Because the platform separates what is being predicted from the information used to make that prediction, three terms are central to the design:~~ graduate.

This section introduces the terms used throughout the paper. ACOSUS treats future transfer-student outcome prediction as a supervised learning problem in which an outcome is predicted from a set of input variables. Because the platform separates what is being measured from the information used to estimate it, three terms are central to the design:

- **Target Survey** / **~~Outcome~~ Readiness Measure (Y)**: The survey instrument that ~~measures what we want to predict, namely~~ produces the ~~student's~~ current ~~likelihood~~ proximal, ~~of~~ self-reported ~~academic success.~~ measure. In machine learning terminology, this is the *label* or *dependent variable*.
- **Factor Survey** / **Predictors (X)**: The survey instruments that collect information used to make predictions, including academic background, financial circumstances, and logistical constraints. In machine learning terminology, these are *features* or *independent variables*.
- **Constructs**: Psychological or educational concepts (e.g., self-efficacy, institutional commitment) that we operationalize through survey questions. A single Factor Survey may measure multiple constructs.

These definitions matter because many of the factors that shape transfer success do not map cleanly onto the information universities already store. Variables such as credit articulation outcomes, transfer shock, belonging, and financial precarity are rarely captured in standard institutional systems (P. M. Nguyen et al., 2025). Advisors also often lack one place to access the information they need; prior work shows that they spend substantial time gathering student data from multiple sources before they can offer useful guidance (Allen et al., 2014). ACOSUS addresses both issues through a dual-survey architecture that supports prediction while also consolidating advisor-facing information.

Prior research identifies several dimensions ~~of~~ associated with transfer success, including academic self-efficacy, institutional commitment, social integration, and career clarity (Wang et al., 2023). Our earlier factor-analysis work grouped related variables into clusters such as academic confidence, time management, financial stability, and support systems (Wang et al., 2023). Building on those findings, in ACOSUS ~~treats~~ the "~~success"~~ readiness ~~as~~ measure" ~~a~~ summarizes ~~student's~~ how ~~likelihood~~ students ~~of~~ assess ~~doing~~ their ~~well~~ present preparation across these ~~dimensions~~ dimensions, ~~and "factors" as~~ while the Factor Surveys capture conditions that may affect ~~that~~ future ~~likelihood.~~ outcomes.

ACOSUS implements this framework through a Dual-Survey Architecture that separates ~~outcome~~ readiness ~~collection~~ measurement from predictor collection (Figure 2). **Target Surveys** (measuring Y) ~~capture~~ ~~success~~ ~~through~~ support two ~~supported~~ modes: (1) a single direct self-assessment question where students rate their ~~success~~ readiness on a 0–100 scale, or (2) a multi-question instrument measuring ~~constructs~~ ~~such~~ ~~as~~ academic confidence, commitment, time management self-efficacy, ~~and~~ career motivation, and related constructs, which are aggregated ~~into~~ ~~a~~ ~~single~~ ~~score~~ ~~via~~ through weighted calculation. **Factor Surveys** (measuring X) collect ~~the~~ predictor variables ~~used~~ ~~for~~ ~~prediction,~~ organized into categories such as academic background, financial circumstances, logistical constraints, and prior interests or experiences.

Beyond their role as predictor vectors for the AI model, Factor Surveys also help advisors by standardizing the collection of transfer-specific information that would otherwise be gathered through lengthy and inconsistent interviews. Every transfer student answers the same set of questions, which supports cohort-level comparison and reduces the chance that important risk factors will be missed. Survey responses are immediately available in the advisor dashboard, which reduces the need to search through emails, notes, or follow-up conversations. The dashboard has been implemented, but its efficiency and usability have not yet been formally evaluated with advisors.

The architecture also supports flexible study designs through **survey linkage**, that is, explicit associations between Target and Factor Surveys. This makes it possible to compare different predictor sets, add new factors over time, and customize the system for different cohorts without changing the core infrastructure. Once these surveys are linked and deployed, the next step is to transform raw responses into model-ready inputs. We return to the linkage mechanism from an implementation perspective when discussing configuration flexibility later in the paper.

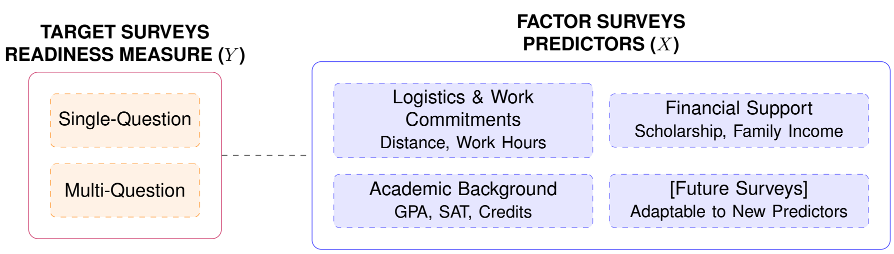

Figure 2: Dual-Survey Architecture. Target Surveys collect the outcome (Y), Factor Surveys collect predictors (X). Dashed lines show linkage relationships enabling flexible study configurations.

Diagram text changes: Single-Question; Multi-Question; Logistics & ~~Commitment~~ Work Commitments Distance, Work Hours; Financial Support Scholarship, Family Income; Academic Background GPA, SAT, Credits; [Future Surveys] Adaptable to New Predictors

Survey Data Processing

The Target and Factor Survey instruments convert student responses into feature vectors for predictive modeling, following the pipeline shown in Figure 3. Each question corresponds to a success-related factor identified in prior research. Our earlier factor-analysis work linked observable indicators such as scholarship status and commute distance to broader constructs such as financial stability and institutional commitment (Wang et al., 2023). The purpose of this stage is to move from survey design to a consistent numerical representation that later modeling stages can use.

Priority Weighting

Not all factors contribute equally to student success. Without an explicit weighting mechanism, a ~~prediction~~ readiness ~~model~~ calculation would treat every survey question as equally informative, giving commute distance the same influence as academic self-efficacy. ~~This~~ ~~contradicts~~ ~~retention~~ Prior research ~~showing~~ suggests that ~~certain~~ some constructs ~~are~~ may ~~substantially~~ be ~~stronger~~ more ~~predictors~~ strongly ~~of~~ associated with persistence than logistical variables (Wang et al., 2023). ~~A~~ An ~~flat~~ equal-weighting ~~weighting~~ scheme would also make the ~~system~~ calculation less interpretable for advisors, who need to understand ~~why~~ how a particular ~~risk~~ score was generated. Prior research ~~consistently~~ identifies ~~some~~ academic, ~~variables~~ financial, ~~academic~~ ~~self-efficacy,~~ ~~financial~~ ~~stability,~~ institutional, and ~~institutional~~ logistical ~~fit,~~ factors ~~for~~ that ~~example~~ can ~~as~~ shape ~~more~~ transfer ~~predictive~~ experiences ~~than~~ ~~others~~ ~~-~~(Ty et al., 2024) +(Ty et al., 2024; Wang et al., 2023). ACOSUS accounts for ~~this~~ these differences through **priority scores**. ~~These~~ The current scores are ~~based~~ literature-informed ~~on~~ starting ~~empirical~~ ~~evidence~~ ~~such~~ ~~as~~ ~~effect~~ ~~sizes~~ values and ~~factor~~ design ~~loadings~~ choices, ~~from~~ not ~~prior~~ coefficients ~~studies,~~ estimated ~~and~~ or ~~they~~ validated using ACOSUS longitudinal outcomes. They can be adjusted as advisor feedback ~~clarifies~~ and institutional evidence clarify which factors ~~matter~~ are most relevant in a ~~given~~ ~~institutional~~ particular context (Table 3).

The priority-weighted aggregation follows a straightforward formulation. For a survey with n questions, where question i has priority score p_i, the student selects an option with weightage w_i^{selected} from a maximum possible w_i^{max}:

$$S = \frac{\sum_{i=1}^{n} \left(\frac{w_i^{\text{selected}}}{w_i^{\text{max}}} \times p_i \right)}{\sum_{i=1}^{n} p_i} \tag{1}$$

This yields a normalized score in $[0, 1]$ that reflects both the selected response and the ~~importance~~ priority of the corresponding question. ~~We~~ A ~~then~~ bounded ~~apply~~ calibration step converts the result to the readiness-score scale. The same priority-weighted calculation can aggregate a ~~logistic~~ multi-question ~~calibration~~ Target ~~curve~~ Survey ~~to~~ and ~~reduce~~ provide ~~overconfident~~ a ~~predictions~~ transparent ~~at~~ summary ~~the~~ of ~~extremes.~~ Factor Survey responses for later exploratory modeling.

~~These~~ To ~~priority~~ show ~~weights~~ how ~~also~~ the ~~remain~~ starting ~~useful~~ priorities affect the result, we recalculated scores for the available deidentified pilot profiles under three alternative scenarios. The analysis used the factor responses represented across those profiles and the same priority-weighted calculation in ~~later~~ each ~~modeling~~ ~~stages.~~ ~~When ACOSUS moves to probabilistic methods, the weights can be used as informative priors or as a principled starting point, so that features with higher priorities have greater initial influence before the model updates them using observed data. This preserves a link between expert knowledge and learned parameters. The next step in processing is to ensure that different kinds of survey responses are normalized appropriately.~~ scenario.

The comparison shows how different design choices change the calculated readiness scores. Equal weighting produced the largest change for one profile, at 5.24 points. These results describe sensitivity within this small set; they do not identify optimal priorities or validate the readiness measure as a predictor of student success. The priorities can be re-estimated and updated as larger samples and longitudinal persistence, credit-accumulation, and completion outcomes become available.

Table 3: ~~Example~~ Literature-informed priority ~~score~~ structure ~~assignments~~ used ~~based~~ in ~~on~~ the ~~literature-informed~~ ~~importance.~~ prototype.

Factor Category	~~Example~~ Question ~~Question~~ Topics	~~Priority~~ ~~Rationale~~ Emphasis
Academic ~~Self-Efficacy~~ background	~~How~~ GPA, ~~confident~~ prior ~~in~~ academic ~~your~~ preparation, ~~ability~~ completed ~~to~~ credits, ~~succeed?~~ and course load	~~9~~ ~~Strong predictor~~ ~~-~~(Bandura, 1997) Higher
~~Financial~~ Program ~~Stability~~ timeline	~~Expected~~ Program ~~financial~~ start ~~stress~~ and ~~impact?~~ expected completion	~~8~~ ~~Affects retention~~ ~~-~~(Vargas et al., 2024) Higher
~~Institutional~~ Financial ~~Commitment~~ support	~~How~~ Scholarship ~~committed~~ and ~~to~~ family ~~completing~~ ~~this~~ ~~program?~~ support	~~9~~ ~~Persistence indicator~~ ~~-~~(Tinto, 1993) Lower
Logistics and commitments	Commute ~~distance~~ and work commitments	Moderate to higher
Interests and experience	~~6~~ Career goals, course interest, and prior experience	~~Practical~~ Varies ~~barrier~~ by factor

Table 4: Priority-weight sensitivity across the available pilot profiles.

Scenario	Mean	Range	Mean $\|\Delta\|$	Max $\|\Delta\|$
Baseline priorities	82.27	72.25–91.38	0.00	0.00
Equal weighting	81.10	67.82–91.16	1.77	5.24
Double academic priorities	81.35	71.27–91.92	1.22	2.63
Double support/logistics priorities	81.76	70.35–91.22	1.23	3.21

These priority weights can also serve as informative priors or principled starting points in later probabilistic models, allowing observed data to update their influence. This preserves a link between prior research, transparent design choices, and learned parameters without treating the starting values as established effects. The next step in processing is to ensure that different kinds of survey responses are normalized appropriately.

Data Type Handling

Survey responses include different data types, and each type requires an appropriate normalization step before modeling, as summarized in Table 5. The distinction between **ordinal** and **cardinal** data is especially important: ordinal responses (e.g., “Not confident” $<$ “Somewhat confident”

$<$ "Very confident") have a meaningful order, whereas cardinal responses (e.g., career aspiration categories) do not imply ranking.

Table 5: Normalization strategies for different data types.

Data Type	Example	Normalization Method
Ordinal	GPA range: "3.0–3.5"	Direct option weightage (e.g., 8/10)
Cardinal	Career: "Industry/Corporate"	Equal weightage or domain-specific mapping
Continuous	SAT score: 1250	Min-max scaling: $(x - \min)/(\max - \min)$

This type-aware processing preserves the meaning of the responses during feature engineering: ordinal relationships remain ordered, nominal categories are not treated as ranked, and continuous values are scaled appropriately. With this processing in place, ACOSUS can pass standardized inputs to the prediction pipeline. The next subsection explains how that pipeline is designed for settings where only a small amount of institutional data is available at first.

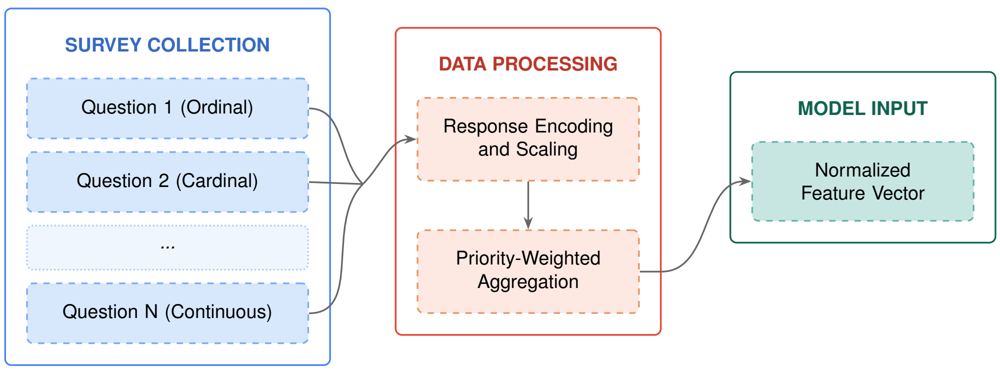

Figure 3: Survey data processing pipeline. Raw responses undergo priority weighting and type-aware normalization to produce standardized feature vectors for model ingestion.

Diagram text changes: Question 1 (Ordinal); Question 2 (Cardinal); ...; Question N (Continuous); SURVEY COLLECTION; ~~Priority~~ Response ~~Weighting;~~ Encoding ~~Type-Aware~~ and ~~Normalization;~~ Scaling; Priority-Weighted Aggregation; DATA PROCESSING; Normalized Feature Vector; MODEL INPUT

Progressive Learning Framework

After survey responses have been standardized, the remaining challenge is how to make use of them when only a small amount of institutional data exists. ACOSUS is designed to work under limited-data conditions from the beginning. Rather than waiting until enough data exist to support prediction, the system provides value from the first student enrollment by giving advisors structured student profiles through standardized surveys. In this way, the early cold-start period still serves a useful advising and data-collection function.

Three-Stage Pipeline

As observations accumulate, the system ~~moves~~ is designed to move through a three-stage pipeline (Figure 4):

Foundation Stage. The system focuses on collecting high-quality labeled observations through surveys that capture both ~~outcomes~~ readiness measure and predictors. Once a minimum corpus exists, lightweight methods (e.g., k-nearest neighbors or cosine similarity matching) can generate initial ~~predictions.~~ estimates. At this stage, data quality matters more than model complexity.

Augmentation Stage. As the corpus grows, the ~~system~~ framework ~~uses~~ can evaluate generative methods (e.g., SMOTE, generative adversarial networks, or variational autoencoders) ~~to~~ for ~~synthesize~~ synthesizing additional training observations. These methods learn from the real data and produce synthetic samples that expand the training set while preserving its main statistical ~~properties.~~ properties; the resulting samples must also be evaluated for fidelity, privacy, bias, and utility.

Refinement Stage. Finally, the ~~system~~ framework ~~transitions~~ can transition to more advanced non-linear models (e.g., deep neural networks, gradient boosting, or ensemble methods) that can capture complex relationships between features and outcomes.

The framework is modular across all three stages. Any compatible predictive method may be used in the foundation stage, any suitable generative technique may be used for augmentation, and any supervised learning architecture may be used in refinement. This design allows researchers to compare algorithms on their own student populations and to adopt new methods as they become available without redesigning the full system. Because those choices may differ across institutions and studies, the next subsection describes the configuration layer that controls how surveys, stages, and algorithms are connected in practice.

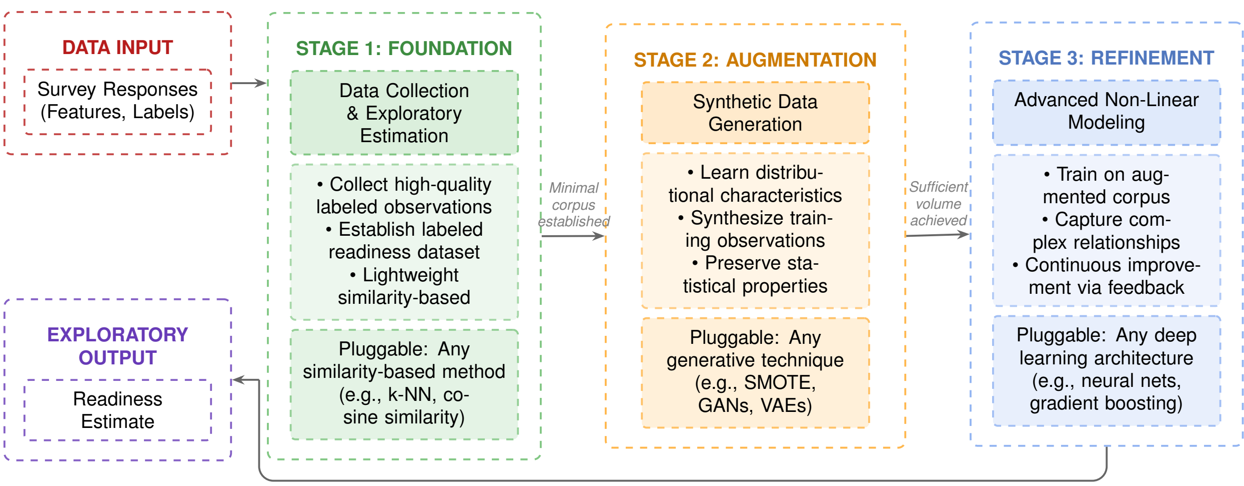

Figure 4: The Three-Stage Progressive Pipeline. Each stage represents a pluggable component that can be substituted with alternative algorithms. The framework progresses from data acquisition through augmentation to refined prediction, with feedback loops enabling continuous improvement.

Diagram text changes: Data Collection & ~~Initial~~ Exploratory ~~Prediction;~~ Estimation; Pluggable: Any similarity-based method (e.g., k-NN, cosine similarity); STAGE 1: FOUNDATION; Synthetic Data Generation; Pluggable: Any generative technique (e.g., SMOTE, GANs, VAEs); STAGE 2: AUGMENTATION; Advanced Non-Linear Modeling; Pluggable: Any deep learning architecture (e.g., neural nets, gradient boosting); STAGE 3: REFINEMENT; DATA INPUT; Survey Responses (Features, Labels); Readiness ~~Score;~~ Estimate; ~~PREDICTION~~ EXPLORATORY OUTPUT

Configuration Flexibility

The modular pipeline design described above is ~~implemented~~ supported ~~through~~ by a configuration system that allows researchers to adapt the framework without modifying the core infrastructure. Four categories of settings ~~support~~ provide this flexibility:

Survey Linkage: The dual-survey architecture uses explicit linkage settings. A single Target Survey (~~outcome~~ readiness measure Y) may be linked to multiple Factor Surveys (predictor sets X_1, X_2, X_3), which allows researchers to examine which predictor combinations perform ~~best.~~ best once suitable longitudinal outcomes exist. Figure 5 shows how linked surveys flow into the model during training and prediction. This arrangement provides several practical advantages:

1. **Comparative predictor analysis:** researchers can deploy alternative Factor Surveys to the same cohort and compare ~~predictive~~ their validity ~~across~~ once ~~predictor~~ suitable ~~sets;~~ outcomes exist;
2. **Longitudinal adaptability:** as research identifies new predictors of transfer success, new Factor Surveys can be added without disrupting existing data collection or invalidating earlier comparisons;
3. **Cohort customization:** different academic programs can deploy program-specific Factor Surveys while maintaining a common outcome measure through shared Target Surveys.

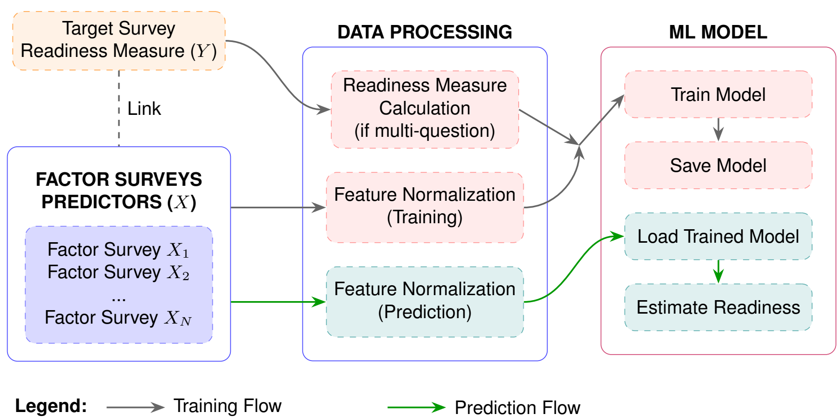

Figure 5: Dual-Survey Architecture with survey linkage. Gray arrows show the Target Survey readiness measure and Factor Survey predictors entering model training, while green arrows show Factor Survey predictors passing through inference. Dashed lines show linkage relationships between Target and Factor Surveys.

> *Diagram text changes:* Target Survey Readiness Measure (~~Label~~ Y); ~~Weighted~~ Readiness Measure Calculation (if multi-question); Feature Normalization (Training); Feature Normalization (Prediction); Factor Survey X_1 Factor Survey X_2 ... Factor Survey X_N; FACTOR SURVEYS PREDICTORS (~~Features~~ X); Train Model; Save Model; Load Trained Model; ~~Predict~~ Estimate Readiness; Legend:

The remaining three categories address the progressive learning pipeline:

Phase Transition Thresholds. Administrators specify how many observations are required before the system moves from one pipeline stage to the next. Default thresholds reflect statistical considerations for each algorithm family, but institutions can adjust them based on enrollment patterns and the desired balance between prediction availability and model reliability.

Algorithm Selection: Each pipeline stage ~~accepts~~ is designed to accept any algorithm that matches the expected interface. The **foundation stage** requires a predictor that works with minimal training data. The **augmentation stage** requires a generative model that can synthesize observations while preserving the main properties of the data. The **refinement stage** accepts supervised learning architectures that can capture non-linear relationships. This modularity allows researchers to compare alternatives on their own student populations rather than rely on fixed defaults.

Hyperparameter Tuning: Within each algorithm, configurable parameters can support fine-tuning for institutional context. Distance metrics, neighbor counts, generation multipliers, network architectures, and training schedules are all exposed as settings. The system logs each configuration together with model performance metrics to support systematic comparison.

This configuration structure ~~allows~~ is intended to allow institutions to adopt newer methods, including improved few-shot learning techniques, stronger data augmentation strategies, or newer model architectures, by updating settings rather than reengineering the full system.

EVALUATION

To assess the usability and perceived value of ACOSUS, we conducted a pilot study with transfer students at a public urban university. This section describes the study design, evaluation instruments, quantitative results, and qualitative findings.

Study Design

Ten transfer students ($N = 10$) from public universities in the United States participated in the study. Participants were recruited by email and enrolled after providing informed consent. All were active transfer students in computing-related majors. The evaluation took place during the foundation phase of the progressive learning pipeline, when the system was still collecting labeled observations for the initial training corpus. At that stage, no predictive model had yet been trained, so the system's main function was structured data collection rather than prediction. Participants completed the Factor and Target Survey instruments, then completed a feedback survey through Qualtrics. The study protocol received Institutional Review Board (IRB) approval before data collection began. The pilot evaluated the student-facing interface and survey workflow, not predictive validity, advisor-dashboard usability, longitudinal outcomes, fairness, or institutional deployment.

Evaluation Instruments

The feedback survey was based on the Technology Acceptance Model (TAM) framework (Davis, 1989) and was used to assess participants' perceptions of the data collection experience. It included 5-point Likert-scale items measuring Perceived Usefulness, Perceived Ease of Use, and Behavioral Intention, along with open-ended questions about interface difficulties, helpful features, missing capabilities, and redesign suggestions. Table 6 presents the constructs and example items.

Table 6: Feedback survey constructs and items.

Construct	Example Item
Perceived Usefulness (PU)	"Using this AI counseling system enables me to accomplish tasks more quickly than other advising methods."
Perceived Ease of Use (PEOU)	"Learning to operate this AI counseling system was easy for me."
Behavioral Intention	"How likely are you to continue using this system for academic planning?"

Open-ended questions captured qualitative feedback on interface difficulties, helpful features, missing capabilities, and redesign suggestions.

Quantitative Results

TAM Construct Scores

The TAM-based measures showed favorable responses on perceived ease of use and perceived usefulness across the participant group (Table 7). Because this evaluation was conducted during the foundation phase, before any predictive output had been generated for participants, the scores reflect participants' experience of the data-collection workflow and interface rather than the system's predictive utility. Lower ratings on usefulness should therefore be read as a reflection of the system's current functionality at the foundation phase rather than as evidence against the underlying design.

Table 7: Summary statistics for evaluation metrics (5-point Likert scale).

Metric	Mean	SD	Min	Max
Perceived Usefulness (PU)	3.37	1.29	1	5
Perceived Ease of Use (PEOU)	4.13	1.05	1	5
Likelihood to Continue	3.12	1.89	1	5
Likelihood to Recommend	3.00	1.77	1	5

Note. Behavioral-intention items have $N = 8$; two participants did not provide a usable rating.

As Figure 6 shows, Perceived Ease of Use received the highest mean among the TAM constructs ($M = 4.13$, $SD = 1.05$), indicating positive agreement that the system is easy to learn and use, with relatively consistent responses across participants. Perceived Usefulness was lower and more dispersed ($M = 3.37$, $SD = 1.29$), suggesting that participants saw the system as easy to operate but were less uniformly convinced of its usefulness for their own advising needs.

Behavioral Intention

Behavioral intention was the weakest dimension in the evaluation (Figure 7). Likelihood to Continue using the system was moderate on average ($M = 3.12$, $SD = 1.89$), and Likelihood to Recommend was slightly lower and similarly dispersed ($M = 3.00$, $SD = 1.77$). Two participants did not provide a usable rating for these items and are excluded from the behavioral-intention means

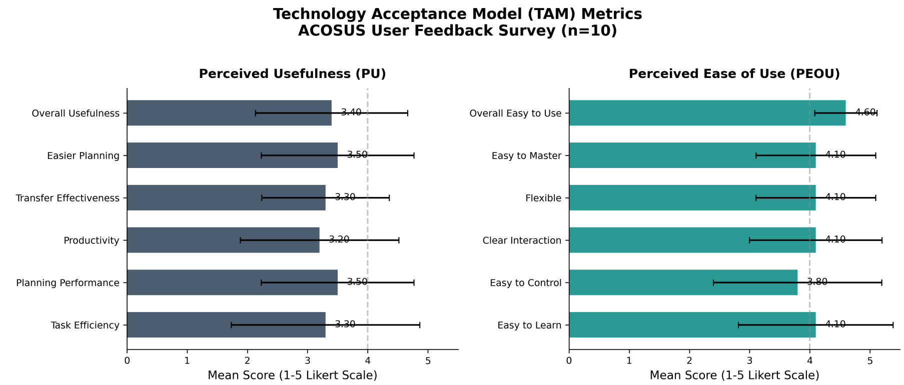

Figure 6: TAM construct scores ($N = 10$). Perceived Ease of Use was the highest-rated TAM construct ($M = 4.13$), while Perceived Usefulness was lower and more dispersed ($M = 3.37$).

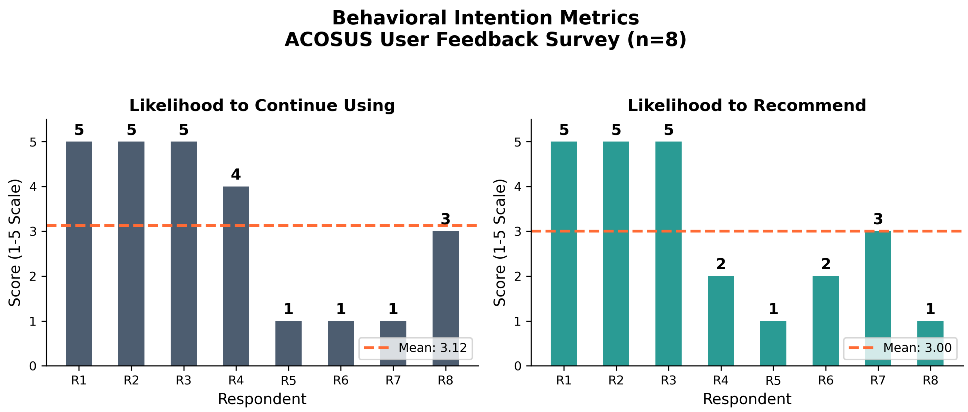

Figure 7: Behavioral intention ($N = 8$). Likelihood to Continue ($M = 3.12$) was slightly higher than Likelihood to Recommend ($M = 3.00$); both showed substantial variance across respondents ($SD \approx 1.8$).

($N = 8$). Both distributions are bimodal: several respondents scored 5/5 while others scored 1–2, with few intermediate ratings. This pattern suggests that the system appeals strongly to a subset of transfer students but not to others, rather than producing uniform mild enthusiasm. We read this cautiously: with only a small number of respondents at the foundation phase, when no predictions had yet been generated, low recommendation scores may reflect limited current utility rather than rejection of the design.

For each respondent, we averaged the available behavioral-intention items. We classified means of 4 or higher as high intention, 2 or lower as low intention, and values between those thresholds as intermediate. One respondent was missing both items. Table +8 reports aggregate means only.

The subgroup averages were consistent with the spread in behavioral intention. High-intention respondents rated both usability and perceived advising value highly. Low-intention respondents still rated ease of use above the scale midpoint but reported lower usefulness, accuracy, alignment, relevance, and actionability. Their open-ended comments emphasized missing course or prerequisite analysis, deeper career exploration, or a preference

Table 8: Exploratory behavioral-intention subgroup means.

Group	N	PU	PEOU	Accuracy	Alignment	Relevance	Actionability
High intention	3	4.83	5.00	4.67	4.67	4.67	5.00
Low intention	4	2.71	3.62	2.25	2.00	2.25	2.50
Intermediate	2	2.92	3.75	4.50	3.00	2.50	3.00
Missing both items	1	2.50	4.33	4.00	–	3.00	2.00

for human advising. Given the very small groups, these comparisons are descriptive only. No significance tests were performed, and the pattern cannot support population-level conclusions.

Qualitative Findings

Open-ended responses yielded five main themes, whose frequencies are shown in Figure 8 and whose representative quotes appear in Table 9.

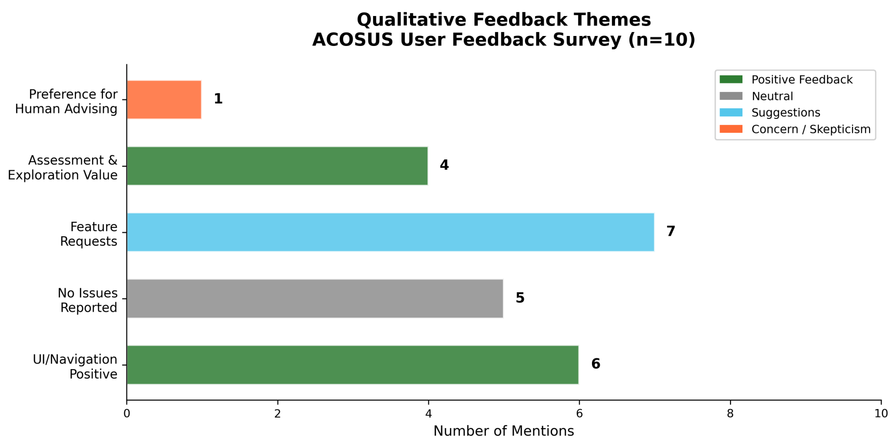

Figure 8: Qualitative theme distribution (N=10). Feature Requests was the most frequent theme (7 mentions), followed by UI/Navigation Positive (6), No Issues Reported (5), Assessment & Exploration Value (4), and a single expression of Preference for Human Advising (1).

Feature Requests was the most frequent theme. Rather than indicating dissatisfaction, the pattern suggests that participants engaged closely enough with the system to want it to do more. Several asked for tighter personalization, with one participant writing that they would "make it tailored more towards individual users and their specific situation," while others requested deeper academic functionality, such as "better analysis of prerequisites to allow for registration" and "more answer choices besides the ones provided." **UI/Navigation Positive** was the next most common theme; participants consistently described the interface as easy to navigate, with one writing that "the UI navigation is very user friendly" and another that "all the interface of the website was very intuitive to interact with," which is consistent with PEOU being the highest-rated TAM construct ($M = 4.13$). **No Issues Reported** formed a third cluster: when asked about missing features or interface difficulties, several participants responded with comments such as "no feature was missing, my experience was great" and "none, everything was clear." Finally, **Assessment & Exploration Value** captured comments in which participants recognized the system's worth for self-assessment and for exploring their options, even at the foundation phase when no predictions

Table 9: Qualitative themes and representative responses ($N = 10$).

Theme	Count	Representative Quote
Feature Requests	7	"Make it tailored more towards individual users and their specific situation"; "Better analysis of prerequisites to allow for registration"
UI/Navigation Positive	6	"The UI navigation is very user friendly"; "All the interface of the website was very intuitive to interact with"
No Issues Reported	5	"No feature was missing, my experience was great"; "None, everything was clear"
Assessment & Exploration Value	4	"The assessment is fun and helpful to me"; "Career exploration"
Preference for Human Advising	1	"I have never had a problem with a human advisor. Please don't replace a human being with AI"

were yet available; participants described it as a tool for "career exploration" and noted that "the assessment is fun and helpful to me." A fifth theme emerged from a single, more skeptical participant, who wrote that they "have never had a problem with a human advisor" and ~~asking~~ asked that the system "not replace a human being with AI." Although isolated, this comment reflects a **Preference for Human Advising** and echoes the advisors' own view that the system should complement rather than substitute for human advising.

Alongside the student pilot, the project also received informal feedback from academic advisors. Advisors agreed that the system addresses a real need and viewed it as a promising direction, while noting that its full value depends on the prediction capability that the foundation phase has not yet produced. They also indicated that the system could help students arrive better prepared for advising meetings, and that consolidating its captured information into a single advising hub would further strengthen its value to advising workflows. These comments motivate formal advisor testing, but they were not collected through a defined advisor sample or usability protocol and are not treated as evaluation evidence.

CONCLUSION AND FUTURE WORK

This paper presented ACOSUS, an AI-driven counseling system designed for transfer students in computing majors. The system addresses two problems that often limit transfer advising: fragmented information and limited institutional data for prediction.

Our primary contributions are: (1) a dual-survey architecture that separates ~~outcome~~ readiness measurement from predictor collection; (2) structured transfer-specific data collection ~~while~~ and ~~providing advisors with structured~~ organized student profiles through a unified advisor dashboard; (~~2~~ 3) a modular prediction architecture with configuration support for survey linkage, phase transitions, and hyperparameters, allowing institutions to adapt the system to their own settings; and (~~3~~ 4) a pilot evaluation with transfer students ($N = 10$) showing that the system is perceived as moderately easy to use and useful, with Perceived Ease of Use ($M = 4.13$)

the strongest of the TAM constructs, alongside notable variance in behavioral intention. The dashboard is implemented, but it was not formally evaluated in the pilot.

The system is currently operational and collecting observations during the foundation phase, but its predictive components still need to be evaluated as the dataset grows. The pilot's small sample size ($N = 10$), small number of participating institutions, voluntary enrollment, and cross-sectional design limit generalizability, and longitudinal outcome tracking was beyond the scope of the present study. Dashboard usability, fairness, and multi-institutional validation also remain to be evaluated.

Future work includes: (1) evaluating the full progressive pipeline once sufficient observations have been collected, including accuracy, calibration, and fairness; (2) deploying the system across multiple institutions to test generalizability; (3) incorporating advisor feedback into the prediction process and evaluating dashboard use in practice; and (4) exploring LLM-based narrative explanations and intervention recommendations.

Taken together, these design choices suggest a practical approach to deploying intelligent advising systems in educational settings where data are limited at the outset. The present pilot supports the design feasibility of the student-facing interface and survey workflow, not predictive validity or improvement in longitudinal student success.

ACKNOWLEDGMENTS

This material is based upon work supported by the National Science Foundation under Grant No. CNS-2219623.

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